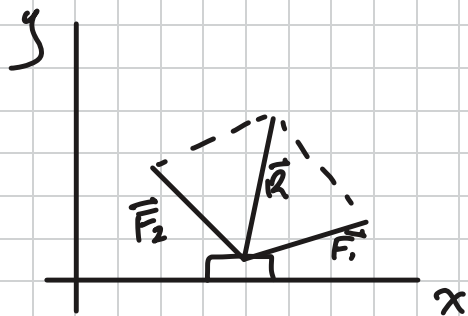


Week 4



$$\vec{R} = \vec{F}_1 + \vec{F}_2$$

resultant or
net force

by components

$$\vec{F}_1 = (F_{1x}, F_{1y}), \quad \vec{F}_2 = (F_{2x}, F_{2y})$$

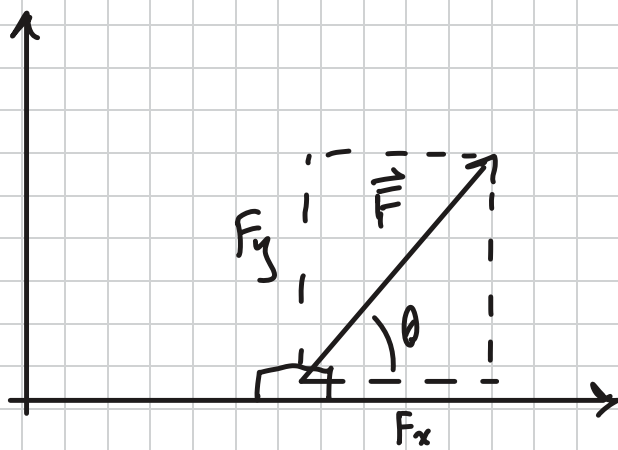
$$\vec{R} = \vec{F}_1 + \vec{F}_2 \quad \vec{R} = (R_x, R_y)$$

$$R_x = F_{1x} + F_{2x}$$

$$R_y = F_{1y} + F_{2y}$$

in general $\vec{R} = \sum \vec{F}$

$$R_x = \sum F_x, \quad R_y = \sum F_y$$



Components

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

Note: while F is positive, the components F_x, F_y can be positive or negative

Newton's first law

An Object on which No external force is acting (or the net external force is zero) will keep its state of rest or keep moving with constant velocity

"rest", "constant velocity" are measured with respect to a frame

Reference frames in which Newton's first law is true are called "inertial reference frames"

$\Rightarrow$ Any frame that moves with constant velocity relative to an inertial frame is also an inertial frame.

Newton's first law is only valid in inertial frames

~~Ex~~ An accelerating vehicle is a non-inertial frame

Newton's Second Law

$$\sum \vec{F} = m \cdot \vec{a}$$

$$\vec{a} = \frac{1}{m} \sum \vec{F} \quad \begin{array}{l} m \text{ mass of} \\ \text{the object} \end{array}$$

Mass is a measure of the inertia of an object.

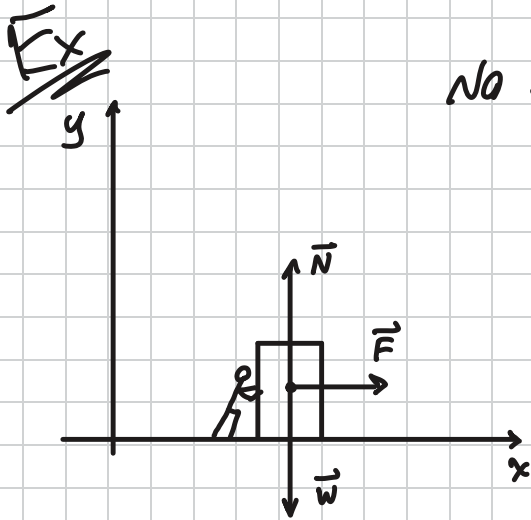
$$\langle m \rangle = \text{kg} \quad \langle a \rangle = \text{m/s}^2$$

$$\langle F \rangle = \text{kg} \cdot \text{m/s}^2 = \text{N}$$

by Component:

$$\sum F_x = ma_x, \quad \sum F_y = ma_y$$

$$\sum F_z = ma_z$$



No friction

$$m = 40 \text{ kg}$$

$$F = 20 \text{ N}$$

What is the resulting acceleration?

$$y: R_y = W_y + N_y = 0$$

$$x: R_x = F = 20 \text{ N}$$

$$a_x = \frac{R_x}{m} = \frac{20 \text{ N}}{40 \text{ kg}} = 0.50 \text{ m/s}^2$$

4.4 mass and weight

$$\vec{F} = m\vec{a} \Rightarrow \vec{a} = \frac{\vec{F}}{m}$$

mass $\rightarrow$ inertia

weight $\rightarrow$ gravitational force

$$\vec{W} = m\vec{g}$$

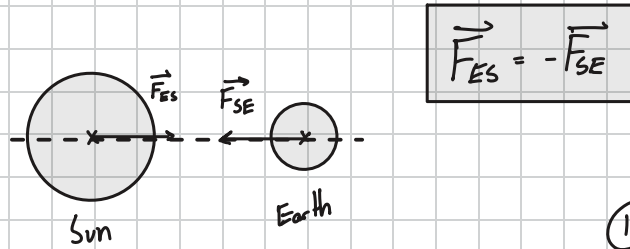
$g = 9.8 \text{ m/s}^2$
at earth surface

$$W = mg$$

if mass = 10 kg $\Rightarrow W = 9.8 \text{ N}$

Newton's third law

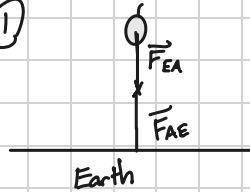
Force is a measure of interaction between two objects



$$\vec{F}_{ES} = -\vec{F}_{SE}$$

Examples

①

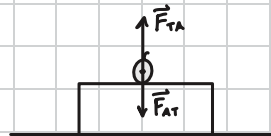


$$F_{EA} = m_A g$$

$$\vec{F}_{AE} = -\vec{F}_{EA}$$

in general $\vec{F}_{12} = -\vec{F}_{21}$
equal magnitude, opposite direction

②

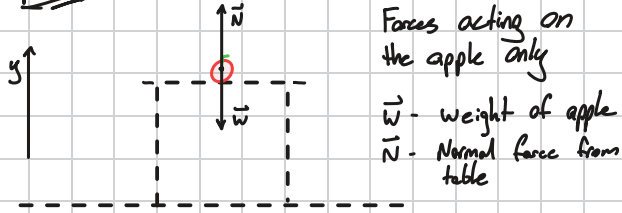


$$\vec{F}_{AT} = -\vec{F}_{TA}$$

Freebody diagrams (FBD)

Draw all forces acting on a specified object

Example



Forces acting on
the apple only
 $\vec{w}$ - weight of apple
 $\vec{N}$ - Normal force from
table

The apple is at rest, so the total force is zero

$$\Rightarrow N - w = 0 \quad \text{but} \quad w = m \cdot g$$

$$\Rightarrow N - m \cdot g = 0 \quad \Rightarrow N = m \cdot g$$

Example

y ↑

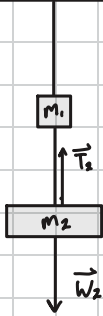


mass m hanging from
the ceiling by a string

$$\vec{T} + \vec{W} = 0 \Rightarrow T - mg = 0$$
$$\Rightarrow T = mg$$

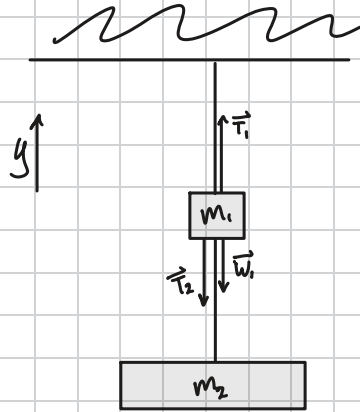
Ex

y ↑



FBD for m_2

$$\vec{T}_2 + \vec{W}_2 = 0$$
$$\vec{T}_2 - m_2 g = 0$$
$$\Rightarrow \boxed{\vec{T}_2 = m_2 g}$$



FBD for m_1

$$\vec{T}_1 + \vec{T}_2 + \vec{W}_1 = 0$$
$$\vec{T}_1 - \vec{T}_2 - m_1 g = 0$$
$$\Rightarrow T_1 - m_2 g - m_1 g = 0$$
$$\Rightarrow T_1 = (m_1 + m_2) g$$

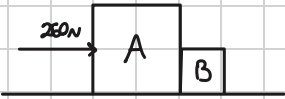
Problem

No friction

$$M_A = 20 \text{ kg}$$

$$M_B = 5 \text{ kg}$$

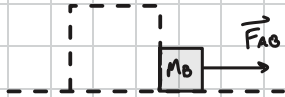
what is the force F_{AB} from A on B



$$M_{A+B} = 25 \text{ kg}, \quad F_{\text{net}} = 250 \text{ N}$$

$$a = \frac{F_{\text{net}}}{M_{A+B}} = \frac{F_{\text{net}}}{M_A + M_B} = \frac{250 \text{ N}}{25 \text{ kg}}$$

$$\Rightarrow a = 10.0 \text{ m/s}^2$$



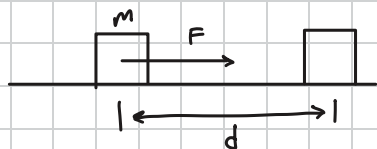
$$F_{AB} = M_B \cdot a$$

$$\Rightarrow F_{AB} = 5.0 \text{ kg} \cdot 10.0 \text{ m/s}^2 = 50 \text{ N}$$

Problem

horz force = 80N moves 11m in 5s

① mass?



$$\Delta t = 5s$$

$$d = 11m$$

$$a = \frac{F}{m}$$

$\Rightarrow$ constant acceleration

$$d = \frac{1}{2} a \cdot (\Delta t)^2 \Rightarrow a = \frac{2d}{(\Delta t)^2}$$

$$a = 0.880 \text{ m/s}^2$$

$$\Rightarrow m = \frac{F}{a} = \frac{80.0 \text{ N}}{0.880 \text{ m/s}^2} \Rightarrow m = 90.9 \text{ kg}$$